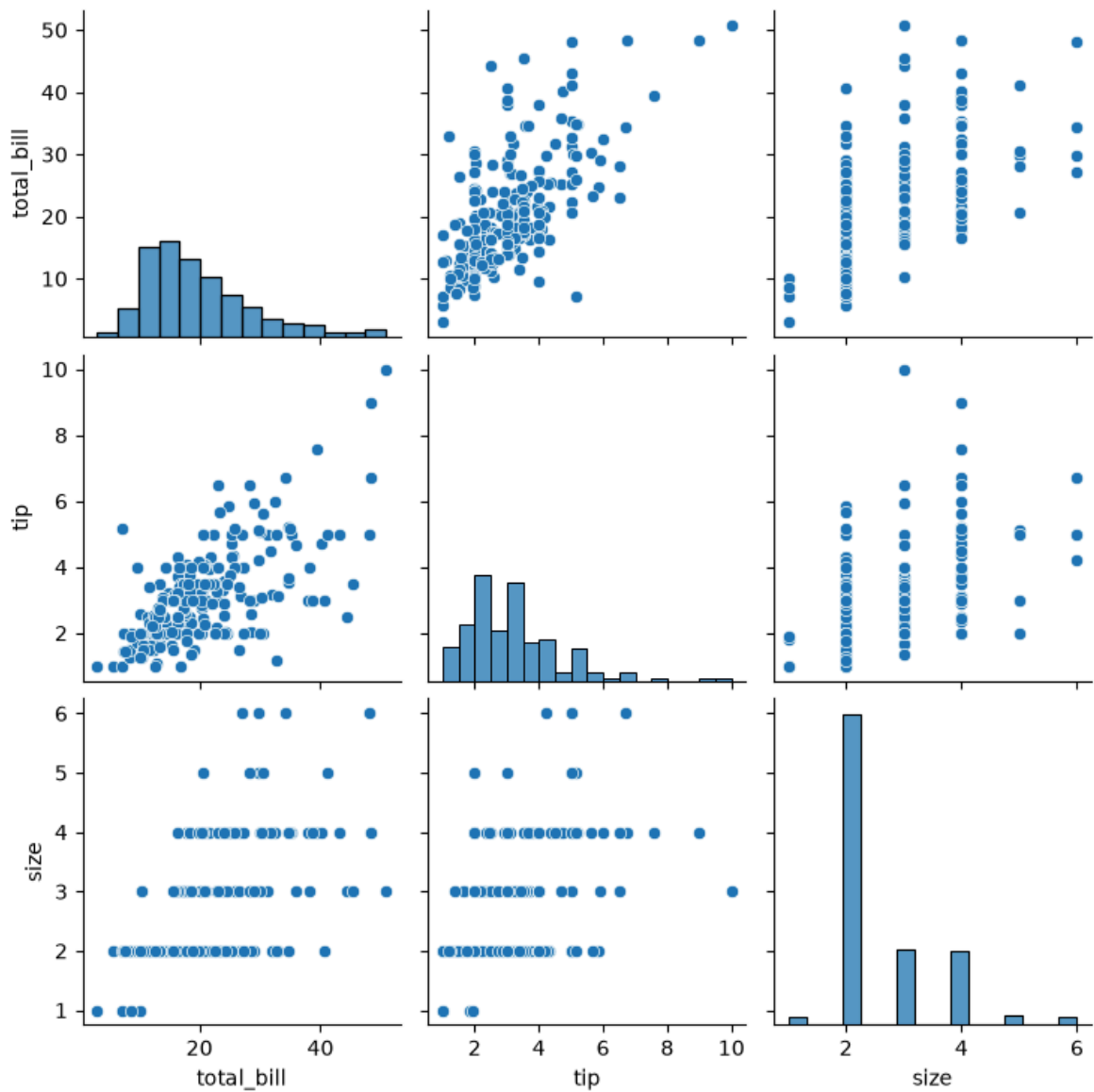


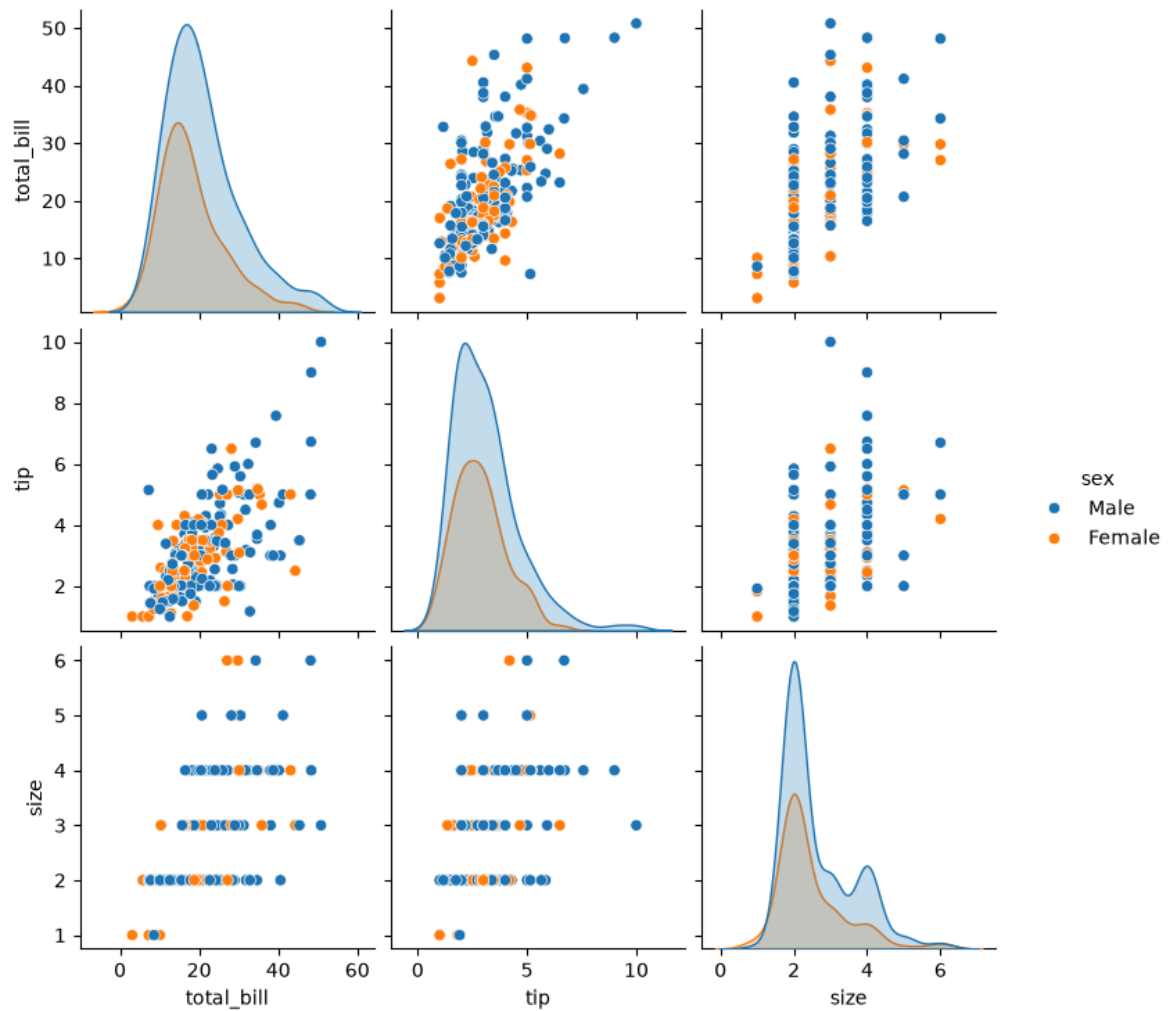
```
In [1]: import seaborn as sns  
import matplotlib.pyplot as plt
```

```
In [2]: tips = sns.load_dataset("tips")
```

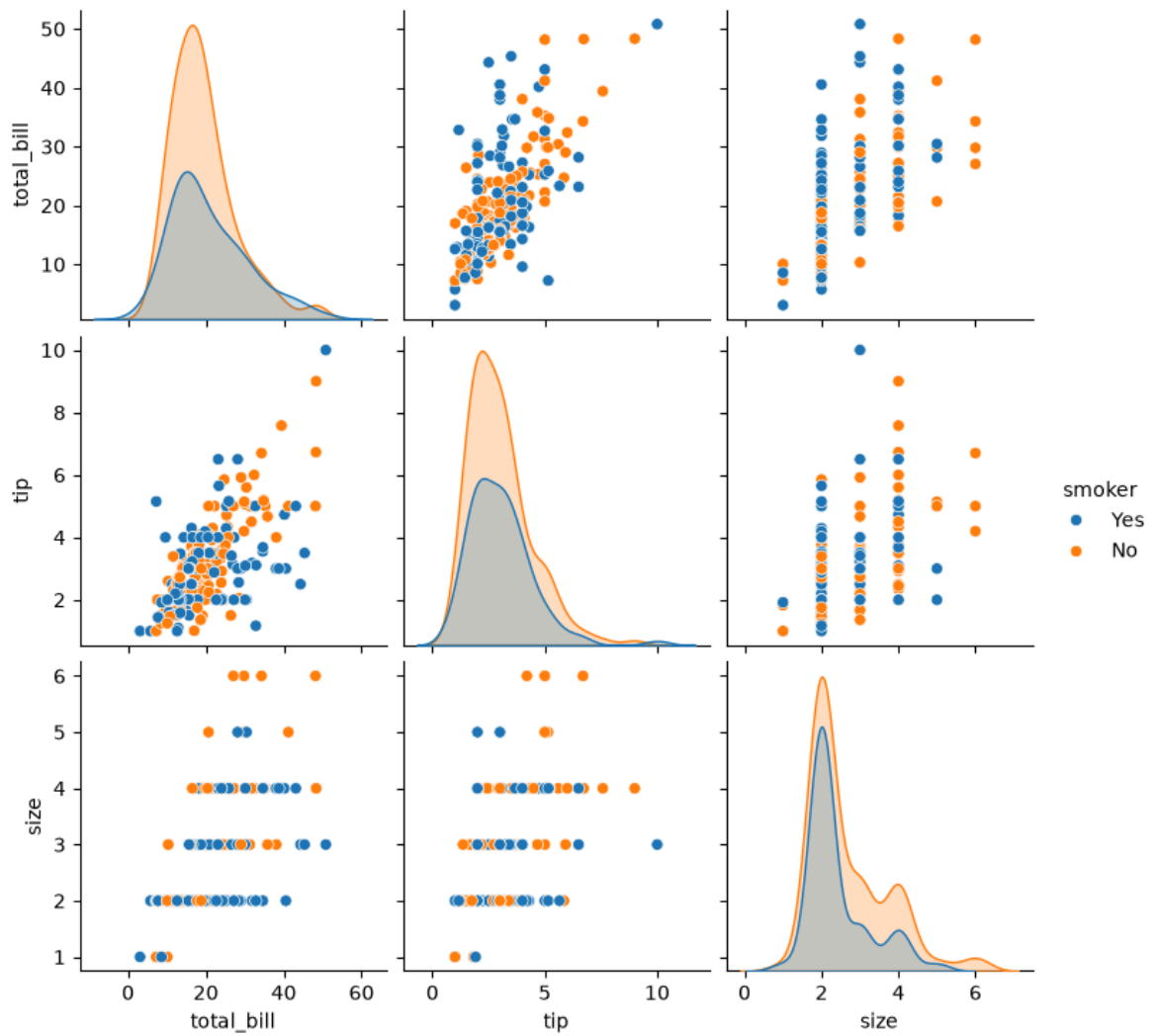
```
In [3]: # TASK 1 - Basic pairplot  
sns.pairplot(tips)  
plt.show()
```



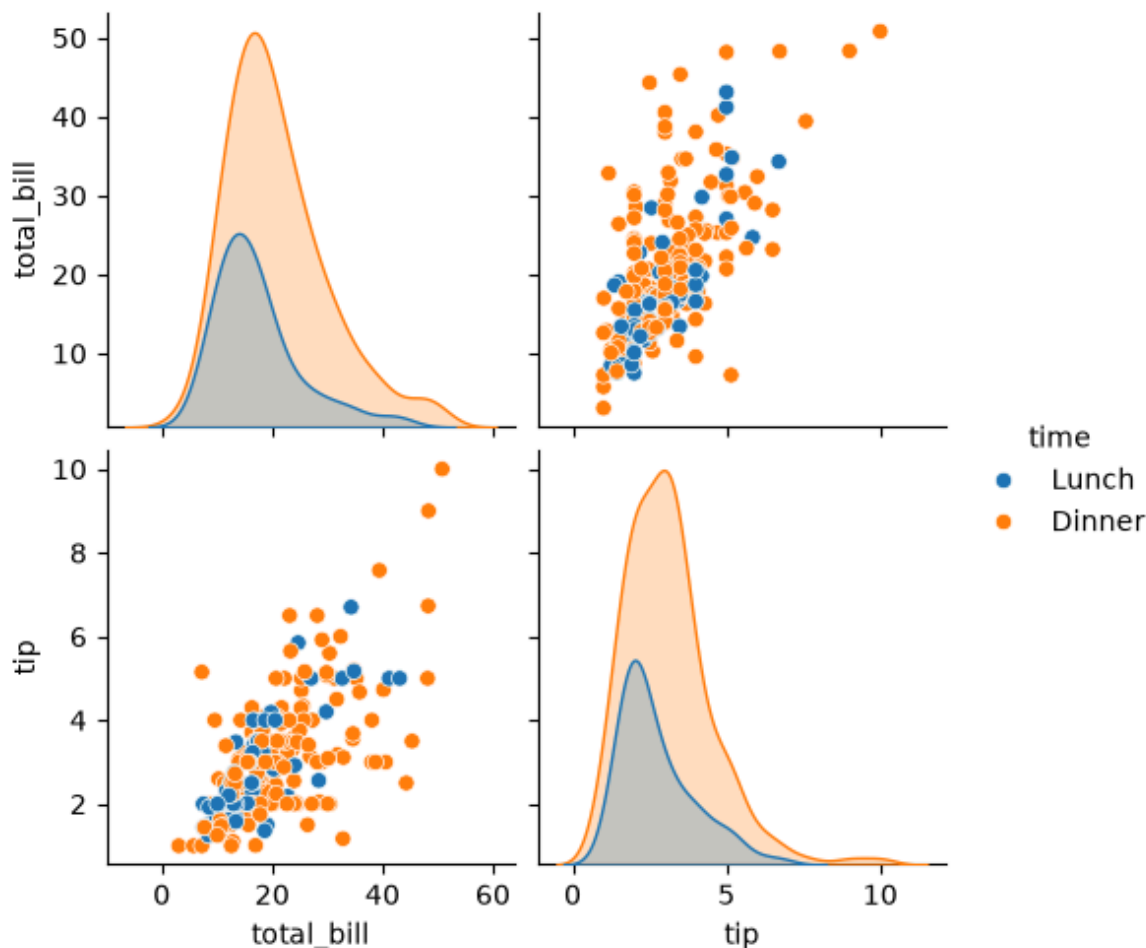
```
In [4]: # TASK 2 - Pairplot with Sex  
sns.pairplot(tips, hue="sex")  
plt.show()
```



```
In [5]: # TASK 3 - Pairplot with Smoker + KDE
sns.pairplot(tips, hue="smoker", diag_kind="kde")
plt.show()
```



```
In [6]: # TASK 4 - Two numerical columns + Time
sns.pairplot(tips, vars=["total_bill", "tip"], hue="time")
plt.show()
```



Analysis Task 1 – Basic Pairplot The pairplot provides an overall view of relationships among the numerical variables.

total_bill and tip show a positive relationship, indicating that larger bills generally correspond to higher tips. Task 2 – Pairplot by Gender Using `hue="sex"` separates observations by gender, making it easier to compare spending and tipping patterns. The distributions show considerable overlap between male and female customers.

Task 3 – Pairplot by Smoking Status The smoker-based visualization helps compare spending and tipping behavior between smokers and non-smokers. The KDE plots on the diagonal clearly show the distribution of each numerical variable for both groups.

Task 4 – Total Bill vs Tip by Time The visualization focuses on total_bill and tip and separates Lunch and Dinner observations. It indicates a positive relationship between bill amount and tip, while Dinner generally contains higher-value bills and tips than Lunch

In []:

In []: